

# ragent Spec Management Manual

This guide explains how to use ragent's built-in spec management system to perform specification-driven design and planning. The `/spec` command family lets you create EARS-notation specifications, generate implementation plans, validate compliance, manage lifecycle status, and drive implementation — all from the interactive TUI.

The commands are exposed via the `/spec` slash command family (TUI) and the `spec_*` LLM tools (`spec_read`, `spec_list`, `spec_search`, `spec_coverage`, `spec_task_update`); the spec lifecycle is managed by the `ragent-specs` crate.

**Scope:** `/spec` slash commands, EARS requirements notation, the SDD (Spec-Driven Development) workflow, lifecycle status transitions, task management, implementation orchestration, JTBD analysis, production feedback, and configuration. For research system commands, see `docs/howtos/research.md`.

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## 1. Purpose and capabilities

The spec management system brings structure to feature design and implementation planning. Instead of ad-hoc prompts, you create formal specifications using EARS (Easy Approach to Requirements Syntax) notation, generate implementation plans with dependency-ordered tasks, validate compliance, and orchestrate implementation.

### What it does

- **EARS specification** — creates `SPEC.md` files using EARS notation with five template types (ubiquitous, event-driven, state-driven, optional, unwanted).
- **Implementation planning** — generates `PLAN.md` with a task table containing IDs, titles, linked requirements, effort estimates, priorities, and dependency chains.
- **Manual test planning** — generates `TESTPLAN.md` with human-readable test cases including pre-conditions, steps, test data, and expected results.
- **EARS validation** — validates spec compliance including section detection, requirement numbering, EARS template matching, ambiguity detection, contradiction detection, and gap analysis.
- **Lifecycle management** — tracks spec status through seven stages with enforced transition rules.
- **Task management** — tracks individual task status (pending, in\_progress, completed, blocked) within a plan.
- **Implementation orchestration** — executes plan tasks in dependency order, injecting per-task prompts into the agent session.
- **JTBD analysis** — performs Jobs-To-Be-Done analysis on existing specs, writing structured job statements with requirement traceability.
- **Production feedback** — appends advisory notes to `FEEDBACK.md` that are surfaced during plan regeneration.
- **SDD workflow separation** — separates specification (`/spec specify`) from planning (`/spec plan`), matching the Spec-Driven Development methodology.
- **Research integration** — links research artifacts to specs via `--from-research`, embedding a `## Related Research` section and YAML frontmatter linkage.
- **Branch-per-spec** — optionally creates a dedicated git branch (`spec/<specname>`) for each specification.

## What it does not do

- It does not auto-generate code — `/spec impl` drives the agent through tasks, but the agent performs the actual implementation.
  - It does not manage CI/CD pipelines — use the VCS tools for that.
  - It does not replace issue tracking — use the GitHub/GitLab tools for external issue management.
- 

## 2. Quick start

Open ragent and create your first spec:

```
/spec create websocket Add a real-time collaborative editing feature using WebSockets
```

The TUI will:

1. Create a new spec directory at `specs/websocket/`.
2. Generate `SPEC.md` (EARS requirements), `PLAN.md` (implementation plan), and `TESTPLAN.md` (manual test plan) using the explore agent.
3. Stream progress to the message window.
4. When finished, validate the spec:

```
/spec validate websocket
```

List all specs:

```
/spec list
```

Check requirement coverage:

```
/spec coverage websocket
```

Start implementation:

```
/spec impl websocket
```

---

## 3. What specs are

A spec is a self-contained directory under `specs/<id>/`:

```
specs/  
  websocket/  
    SPEC.md          # EARS requirements specification  
    PLAN.md          # implementation plan with task table  
    TESTPLAN.md      # manual test plan with test cases  
    TASKS.md         # ordered task list (from /spec tasks)  
    quickstart.md     # quickstart validation scenarios (from /spec tasks)  
    JTBD.md          # Jobs-To-Be-Done analysis (from /spec jtbd)  
    FEEDBACK.md       # production feedback notes (from /spec feedback)  
    data-model.md     # data entity model (from /spec plan, if enabled)  
    contracts/        # API contract definitions (from /spec plan, if enabled)  
  auth-refactor/  
    SPEC.md
```

PLAN.md  
TESTPLAN.md

### File descriptions

| File          | Created by                             | Description                                               |
|---------------|----------------------------------------|-----------------------------------------------------------|
| SPEC.md       | /spec create, /spec specify            | EARS requirements specification with YAML frontmatter     |
| PLAN.md       | /spec create, /spec plan, /spec update | Implementation plan with task table                       |
| TESTPLAN.md   | /spec create, /spec update             | Manual test plan with test cases                          |
| TASKS.md      | /spec tasks                            | Ordered task list derived from PLAN.md                    |
| quickstart.md | /spec tasks                            | Quickstart validation scenarios from SPEC.md requirements |
| JTBD.md       | /spec jtbd                             | Jobs-To-Be-Done analysis with coverage matrix             |
| FEEDBACK.md   | /spec feedback                         | Production feedback notes (advisory)                      |
| data-model.md | /spec plan (if sdd.data_model enabled) | Data entity model with relationships and constraints      |
| contracts/    | /spec plan (if sdd.contracts enabled)  | API contract definitions                                  |

### Naming rules

Spec IDs must be valid directory names: alphanumeric characters, hyphens, and underscores. The SpecId type validates this at creation time.

Valid IDs: websocket, auth-refactor, q3-review, api-v2, auth\_refactor

Invalid IDs: WebSocket (uppercase), api.v2 (dots)

## 4. The EARS notation

EARS (Easy Approach to Requirements Syntax) is a structured template for writing unambiguous requirements. ragent's spec system uses five EARS template types:

| Template            | Pattern                                              | Use case                       |
|---------------------|------------------------------------------------------|--------------------------------|
| <b>Ubiquitous</b>   | The <system> shall <response>.                       | Always-on requirements         |
| <b>Event-driven</b> | When <trigger>, the <system> shall <response>.       | Reactive requirements          |
| <b>State-driven</b> | While <precondition>, the <system> shall <response>. | Conditional state requirements |

| Template        | Pattern                                                     | Use case                   |
|-----------------|-------------------------------------------------------------|----------------------------|
| <b>Optional</b> | Where <feature> is included, the <system> shall <response>. | Feature-gated requirements |
| <b>Unwanted</b> | If <trigger>, the <system> shall <response>.                | Error/exception handling   |

Each spec should use at least one of each template type. Requirements are numbered as FR-001, FR-002, etc. (functional requirements) or NFR-001, NFR-002, etc. (non-functional requirements).

### Example SPEC.md structure

```

---
status: draft
audit:
  - { time: 1787099746, from: "none", to: "draft", actor: "system" }
---

# Specification: websocket - Real-time Collaborative Editing

## Overview

This specification defines a WebSocket-based real-time collaborative editing
feature for the application.

## Requirements

### FR-001
The system shall establish a WebSocket connection within 2 seconds of user
action. (Ubiquitous)

### FR-002
When a user edits a document, the system shall broadcast the change to all
connected clients within 100ms. (Event-driven)

### FR-003
While the connection is unstable, the system shall queue edits and replay
them when connectivity is restored. (State-driven)

### FR-004
Where CRDT support is included, the system shall merge concurrent edits
without conflict. (Optional)

### FR-005
If the WebSocket server is unreachable, the system shall display a
reconnection prompt with a retry button. (Unwanted)

```

---

## 5. The `/spec` command family

All commands are entered in the TUI input box.

### 5.1 `/spec help`

Show the command reference table with all subcommands, arguments, and descriptions.

```
/spec help
```

### 5.2 `/spec create <specname> <feature> [--from-research <name>]`

Create a new spec with `SPEC.md`, `PLAN.md`, and `TESTPLAN.md` in one step. This is the all-in-one command for getting a spec off the ground quickly.

The `--from-research <name>` flag links a research artifact by adding a `research:` field to the YAML frontmatter and a `## Related Research` section to the `SPEC.md` body.

```
/spec create websocket Add a real-time collaborative editing feature using WebSockets
/spec create auth-v2 Implement OAuth2 authentication with PKCE flow
/spec create data-export Add CSV and JSON export for all list views --from-research export-format
```

### 5.3 `/spec specify <specname> <feature> [--from-research <name>]`

Create a `SPEC.md` only — no `PLAN.md` or `TESTPLAN.md`. This separates the specification stage from the planning stage, matching the SDD workflow. The agent inserts `[NEEDS CLARIFICATION: <question>]` markers wherever a requirement is ambiguous.

After reviewing and resolving clarification markers, use `/spec plan` to generate the implementation plan.

When `sdd.branch_per_spec` is enabled in `ragent.json`, this command also creates a `spec/<specname>` git branch.

```
/spec specify websocket Add a real-time collaborative editing feature
/spec specify auth-v2 Implement OAuth2 with PKCE --from-research oauth-study
```

### 5.4 `/spec plan <spec-id> <tech-context>`

Generate or regenerate `PLAN.md` from an existing `SPEC.md` using the provided technology context as guidance. This is the second stage of the SDD workflow after `/spec specify`.

The technology context is a free-text description that informs technology choices and rationale documented in the plan. If a `PLAN.md` already exists, task statuses for unchanged task IDs are preserved.

When `sdd.data_model` is enabled, the prompt also instructs the agent to generate an optional `data-model.md` artifact. When `sdd.contracts` is enabled, it instructs the agent to generate an optional `contracts/` directory. When `FEEDBACK.md` exists, its notes are surfaced in the prompt.

```
/spec plan websocket Use tokio-tungstenite for WebSocket server, Redis for pub/sub
/spec plan auth-v2 Use oauth2 crate with PKCE, store tokens in SQLite via rusqlite
/spec plan api-v2 Migrate from REST to gRPC using tonic, protobuf schemas in proto/ directory
```

### 5.5 /spec update <spec-id>

Regenerate PLAN.md and TESTPLAN.md from an edited SPEC.md. This command re-reads the current SPEC.md (which you may have manually edited) and regenerates the plan and test plan to match. It does NOT modify SPEC.md.

Task statuses for unchanged task IDs are preserved.

```
/spec update websocket  
/spec update auth-v2
```

### 5.6 /spec add <spec-id> <feature>

Incrementally add requirements to an existing spec and update its plan. The agent reads the existing SPEC.md and PLAN.md, generates incremental requirements and tasks for the new feature, and inserts them without modifying existing sections.

```
/spec add websocket Add presence indicators showing who is currently viewing each document  
/spec add auth-v2 Add refresh token rotation with sliding expiration
```

### 5.7 /spec validate [spec-id]

Validate EARS compliance. Without a spec ID, validates all specs.

Validation checks include:

- **Required section detection** — verifies ## Requirements section exists
- **Requirement numbering** — checks for gaps (e.g. FR-001, FR-003 missing FR-002) and duplicates
- **EARS template matching** — verifies each requirement matches a known EARS pattern
- **Status value validation** — checks the YAML frontmatter status is a valid SpecStatus
- **PLAN.md completeness** — verifies the plan has a ## Tasks table
- **Clarification markers** — detects [NEEDS CLARIFICATION] markers (when `sdd.clarification_markers` is enabled)
- **Ambiguity detection** — flags vague terms, undefined acronyms (when `sdd.consistency_checks` is enabled)
- **Contradiction detection** — finds opposing requirement claims (when `sdd.consistency_checks` is enabled)
- **Gap analysis** — identifies missing testable criteria (when `sdd.consistency_checks` is enabled)
- **Phase-1 gates** — validates pre-implementation gate checkboxes (when `sdd.phase_minus_one_gates` is enabled)

```
/spec validate websocket  
/spec validate          # validate all specs
```

### 5.8 /spec list [--status <status>] [--prefix <prefix>]

List all specs with optional filtering.

Filters:

- `--status <status>` — filter by lifecycle status (draft, in\_review, approved, in\_progress, implemented, verified, archived)
- `--prefix <prefix>` — filter by spec ID prefix (case-insensitive)
- `--all` — include archived specs (excluded by default)

By default, specs are sorted by most recently modified. Other sort options: by status (lifecycle order), by ID (alphanumeric), or by title.

```
/spec list
/spec list --status approved
/spec list --prefix auth
/spec list --all
```

### 5.9 /spec search <query>

Full-text search across all specs. Returns matching spec IDs, titles, and context snippets around the matches.

```
/spec search websocket
/spec search authentication OAuth
/spec search "error handling"
```

### 5.10 /spec status <spec-id> [<new-status>]

Show the current status of a spec, or transition it to a new status.

Status transitions are enforced — you cannot skip stages. See section 6 for the full transition graph.

```
/spec status websocket                # show current status
/spec status websocket in_review      # transition to in_review
/spec status websocket approved       # transition to approved
```

### 5.11 /spec task <spec-id> [<task-id>] [<new-status>]

List tasks, show a single task, or update a task's status.

```
/spec task websocket                # list all tasks
/spec task websocket T-003           # show task T-003
/spec task websocket T-003 in_progress # mark T-003 as in_progress
/spec task websocket T-003 completed  # mark T-003 as completed
/spec task websocket T-005 blocked    # mark T-005 as blocked
```

### 5.12 /spec activate <spec-id>

Activate a spec for context injection. When a spec is active, its SPEC.md content is injected into the agent's system prompt so the agent follows the spec's requirements during implementation.

```
/spec activate websocket
```

### 5.13 /spec deactivate

Deactivate the currently active spec. No more spec context will be injected into agent prompts.

```
/spec deactivate
```

### 5.14 /spec coverage <spec-id>

Show a requirement coverage report. This displays which requirements have linked tasks and which do not, helping identify gaps in the implementation plan.

```
/spec coverage websocket
```

### 5.15 /spec impl <spec-id> [--task <ID>] [--dry-run]

Implement a spec by executing its PLAN.md tasks in dependency order. The runner reads the plan, resolves dependency order, and constructs per-task prompts that drive the agent through implementing each task sequentially.

Flags:

- **--task <ID>** — execute only the specified task and its transitive dependencies
- **--dry-run** — display the execution plan without actually running tasks

Alias: /spec implement

```
/spec impl websocket                # implement all tasks in order
/spec impl websocket --dry-run      # preview the execution plan
/spec impl websocket --task T-003   # implement only T-003 and its deps
/spec implement auth-v2             # alias
```

### 5.16 /spec tasks <spec-id>

Generate TASKS.md containing an ordered task list derived from the existing PLAN.md, plus quickstart.md with key validation scenarios derived from SPEC.md requirements.

```
/spec tasks websocket
```

### 5.17 /spec jtbd <spec-id> [--force] [--agent <name>]

Perform a Jobs-To-Be-Done analysis of an existing spec's SPEC.md, writing the result to specs/<spec-id>/JTBD.md.

The analysis extracts jobs using the JTBD framework:

- **Job statement** — “When <situation>, I want to <motivation>, so I can <expected outcome>.”
- **Job type** — functional, emotional, or social
- **Performer** — who is hiring the product
- **Related requirements** — FR/NFR IDs that trace to this job
- **Success signals** — observable indicators the job is being fulfilled

The output also includes an out-of-scope section and a coverage matrix mapping requirements to jobs.

Flags:

- **--force** — overwrite an existing JTBD.md
- **--agent <name>** — dispatch to a specific agent (default: explore agent)

```
/spec jtbd websocket
/spec jtbd websocket --force
/spec jtbd auth-v2 --agent architect
```



### 5.18 /spec feedback <spec-id> <note>

Append a production feedback note to `specs/<spec-id>/FEEDBACK.md`. Notes are advisory — they are surfaced during `/spec plan` regeneration but do not block validation or status transitions.

Use this to capture production metrics, incident reports, or user feedback that should inform future plan iterations.

```
/spec feedback websocket Users report intermittent disconnections when switching networks
```

```
/spec feedback auth-v2 Token refresh fails silently when the server clock drifts by more than 5 s
```

### 5.19 /spec delete <spec-id> [--yes]

Delete a spec directory from the workspace. The `--yes` flag skips the confirmation prompt.

```
/spec delete websocket --yes
```

```
/spec delete old-feature --yes
```

---

## 6. Lifecycle status and transitions

Specs progress through seven lifecycle stages. Transitions are enforced — you cannot skip stages.

### Status values

| Status                   | Description                            |
|--------------------------|----------------------------------------|
| <code>draft</code>       | Initial draft state                    |
| <code>in_review</code>   | Under review by stakeholders           |
| <code>approved</code>    | Approved and ready for implementation  |
| <code>in_progress</code> | Implementation in progress             |
| <code>implemented</code> | Code implementation complete           |
| <code>verified</code>    | Verified as correct and complete       |
| <code>archived</code>    | Archived, excluded from active queries |

### Transition graph

```
draft → in_review → approved → in_progress → implemented → verified → archived
      ↑                                     |
```

(archived → draft)

| From                     | Allowed transitions                        |
|--------------------------|--------------------------------------------|
| <code>draft</code>       | <code>in_review</code>                     |
| <code>in_review</code>   | <code>draft</code> , <code>approved</code> |
| <code>approved</code>    | <code>in_progress</code>                   |
| <code>in_progress</code> | <code>implemented</code>                   |
| <code>implemented</code> | <code>verified</code>                      |
| <code>verified</code>    | <code>archived</code>                      |
| <code>archived</code>    | <code>draft</code> (reactivate)            |

| From | Allowed transitions |
|------|---------------------|
|------|---------------------|

Note: `in_review` can go back to `draft` (changes requested) or forward to `approved` (sign-off). Archived specs can be reactivated back to `draft`.

### Transition gating

When SDD flags are enabled, additional gates may block transitions:

- **Clarification markers** — if `sdd.clarification_markers` is enabled, transitioning to `approved` is blocked when unresolved `[NEEDS CLARIFICATION]` markers exist in the spec.
- **Phase -1 gates** — if `sdd.phase_minus_one_gates` is enabled, transitioning to `in_progress` is blocked until all pre-implementation gate checkboxes in `PLAN.md` are checked.

## 7. Task management

Tasks within a `PLAN.md` have their own status tracking.

### Task status values

| Status                   | Description                   |
|--------------------------|-------------------------------|
| <code>pending</code>     | Task not yet started          |
| <code>in_progress</code> | Task actively being worked on |
| <code>completed</code>   | Task completed                |
| <code>blocked</code>     | Task blocked or deferred      |

### Task table structure

The `PLAN.md` task table uses these columns:

| Column       | Description                                                                                |
|--------------|--------------------------------------------------------------------------------------------|
| ID           | Task identifier (T-001, T-002, ...)                                                        |
| Title        | Human-readable task title                                                                  |
| Requirement  | Linked requirement IDs (FR-007, NFR-003, ...)                                              |
| Effort       | Size estimate: S, M, or L                                                                  |
| Priority     | Critical, High, Medium, or Low                                                             |
| Status       | Task status: Pending, <code>in_progress</code> , completed, or blocked (initially Pending) |
| Dependencies | Task IDs that must complete first                                                          |

### Dependency ordering

The implementation runner (`/spec impl`) resolves dependencies topologically and executes tasks in order. Tasks with no dependencies run first; tasks blocked by incomplete dependencies are deferred.

## 8. The SDD workflow

The Spec-Driven Development (SDD) workflow separates specification from planning, allowing for review and clarification between stages.

### SDD workflow stages

1. `/spec specify <name> <feature>`  
↓  
SPEC.md created with [NEEDS CLARIFICATION] markers  
↓
2. Review SPEC.md, resolve clarification markers  
↓
3. `/spec validate <name>`  
↓  
Validation passes  
↓
4. `/spec plan <name> <tech-context>`  
↓  
PLAN.md created from SPEC.md + technology context  
↓
5. `/spec tasks <name>`  
↓  
TASKS.md + quickstart.md created  
↓
6. `/spec status <name> approved`  
↓
7. `/spec impl <name>`  
↓  
Tasks executed in dependency order

### SDD vs. quick-create workflow

**Quick-create** (`/spec create`) generates everything in one step — ideal for small features where the specification is straightforward and the technology choices are obvious.

**SDD workflow** (`/spec specify` then `/spec plan`) separates the stages — ideal for larger features where requirements need review and clarification before planning, or where technology choices need careful consideration.

### `/spec plan` vs `/spec update`

Both regenerate PLAN.md, but they differ:

- `/spec plan <id> <tech-context>` — generates the plan from SPEC.md using a technology context argument. Use this when you want to guide technology choices. This is the SDD stage-2 command.
- `/spec update <id>` — regenerates the plan from an edited SPEC.md without a technology context. Use this after manually editing SPEC.md to add or modify requirements. Also regenerates TESTPLAN.md.

---

## 9. SDD configuration

SDD capabilities are gated by configuration flags in `ragent.json` under the `sdd` key. All flags default to `false` (opt-in).

```
{
  "sdd": {
    "clarification_markers": true,
    "quality_checklists": true,
    "constitution": false,
    "phase_minus_one_gates": true,
    "branch_per_spec": true,
    "research_artifacts": true,
    "data_model": true,
    "contracts": true,
    "quickstart": true,
    "test_first_ordering": false,
    "consistency_checks": true,
    "amendment_process": false,
    "feedback_loop": true
  }
}
```

### Flag reference

| Flag                               | Effect                                                                                            |
|------------------------------------|---------------------------------------------------------------------------------------------------|
| <code>clarification_markers</code> | Enable [NEEDS CLARIFICATION] marker detection in validation                                       |
| <code>quality_checklists</code>    | Embed quality checklists in spec and plan templates                                               |
| <code>constitution</code>          | Generate and parse CONSTITUTION.md architectural-principles artifact                              |
| <code>phase_minus_one_gates</code> | Enable Phase -1 pre-implementation gate validation                                                |
| <code>branch_per_spec</code>       | Create a git branch ( <code>spec/&lt;specname&gt;</code> ) per spec on <code>/spec specify</code> |
| <code>research_artifacts</code>    | Link research artifacts into SPEC.md frontmatter                                                  |
| <code>data_model</code>            | Generate <code>data-model.md</code> during <code>/spec plan</code>                                |
| <code>contracts</code>             | Generate <code>contracts/</code> directory during <code>/spec plan</code>                         |
| <code>quickstart</code>            | Generate <code>quickstart.md</code> validation scenarios                                          |
| <code>test_first_ordering</code>   | Enforce test-first file creation ordering in plans                                                |
| <code>consistency_checks</code>    | Run ambiguity, contradiction, and gap consistency checks                                          |
| <code>amendment_process</code>     | Enable constitutional amendment process with dated changelog                                      |
| <code>feedback_loop</code>         | Enable production feedback loop ( <code>FEEDBACK.md</code> surfacing)                             |

---

## 10. Validation details

`/spec validate` runs a series of checks on `SPEC.md` (and optionally `PLAN.md`). Issues are categorised by severity and category.

### Severity levels

| Severity | Description                         |
|----------|-------------------------------------|
| Error    | Must fix — blocks progression       |
| Warning  | Should fix — may indicate a problem |
| Info     | Informational — no action required  |

### Issue categories

| Category          | Description                                                      |
|-------------------|------------------------------------------------------------------|
| EarsSyntax        | Requirement does not match any EARS template pattern             |
| MissingSection    | Required section (e.g. <code>## Requirements</code> ) is missing |
| InvalidStatus     | YAML frontmatter status is not a valid <code>SpecStatus</code>   |
| Numbering         | Requirement numbering has gaps or duplicates                     |
| Plan              | <code>PLAN.md</code> is missing or incomplete                    |
| Structure         | Spec structure does not follow expected layout                   |
| Clarification     | Unresolved <code>[NEEDS CLARIFICATION]</code> markers found      |
| Ambiguity         | Vague terms or undefined acronyms detected                       |
| Contradiction     | Opposing requirement claims detected                             |
| Gap               | Missing testable criteria or coverage gaps                       |
| PhaseMinusOneGate | Pre-implementation gate checkboxes not checked                   |

### Validation categories gated by SDD flags

When the corresponding SDD flag is `false`, the check is skipped:

| Check                   | Gated by                               |
|-------------------------|----------------------------------------|
| Clarification markers   | <code>sdd.clarification_markers</code> |
| Ambiguity detection     | <code>sdd.consistency_checks</code>    |
| Contradiction detection | <code>sdd.consistency_checks</code>    |
| Gap analysis            | <code>sdd.consistency_checks</code>    |
| Phase -1 gates          | <code>sdd.phase_minus_one_gates</code> |
| Constitution            | <code>sdd.constitution</code>          |

## 11. Implementation orchestration

`/spec impl` reads the `PLAN.md`, parses the task table, resolves dependency order, and drives the agent through each task.

## How it works

1. **Parse** — reads `PLAN.md` and extracts the task table into structured `PlanTask` objects with IDs, titles, requirements, effort, priority, and dependencies.
2. **Resolve order** — topologically sorts tasks by dependencies. Tasks with no dependencies run first; tasks blocked by incomplete dependencies are deferred.
3. **Build prompts** — constructs a per-task prompt that includes the task title, description, linked requirements, and the spec's `SPEC.md` content.
4. **Execute** — injects each task prompt into the agent session one at a time. The agent implements the task, and the runner moves to the next.
5. **Track progress** — emits progress updates showing completed, skipped, and blocked task counts.

## Dry-run mode

`--dry-run` displays the execution plan without running any tasks. This shows the resolved execution order, milestone groups, and effort summary.

```
/spec impl websocket --dry-run
```

## Single-task mode

`--task <ID>` executes only the specified task and its transitive dependencies. This is useful for re-running a failed task or implementing a specific subset.

```
/spec impl websocket --task T-003
```

## Milestone groups

The runner identifies milestone groups from the plan — clusters of tasks that share a milestone deliverable. This helps track progress at a higher level than individual tasks.

---

## 12. Research integration

Specs can be linked to research artifacts using `--from-research`. This creates a two-way linkage:

1. The `SPEC.md` YAML frontmatter includes a `research:` field naming the research artifact.
2. A `## Related Research` section is added to the `SPEC.md` body.
3. The research artifact's `INDEX.md` is updated to reference the spec.

## Linking a spec to research

```
/spec create websocket Add real-time collaborative editing --from-research realtime-collab
/spec specify auth-v2 Implement OAuth2 with PKCE --from-research oauth-study
```

## Configuration

The `sdd.research_artifacts` flag in `ragent.json` enables this feature:

```
{
  "sdd": {
    "research_artifacts": true
  }
}
```

```
}  
}
```

---

### 13. Branch-per-spec

When `sdd.branch_per_spec` is enabled, `/spec specify` creates a dedicated git branch named `spec/<specname>` to isolate spec work.

```
{  
  "sdd": {  
    "branch_per_spec": true  
  }  
}
```

The branch creation is tolerant — if the working directory is not a git repository or the branch already exists, spec creation proceeds without branching.

---

### 14. JTBD analysis

`/spec jtbd` performs a Jobs-To-Be-Done analysis on an existing spec, extracting the underlying “jobs” the feature is hired to do.

#### Output structure

The `JTBD.md` file contains:

1. **YAML frontmatter** with `status: draft`
2. **Overview** — one or two sentence summary of the spec
3. **Jobs** — each job with:
  - Job statement using the grammar: “When <situation>, I want to <motivation>, so I can <expected outcome>.”
  - Job type: functional, emotional, or social
  - Performer: who is hiring the product
  - Related requirements: FR/NFR IDs that trace to this job
  - Success signals: observable indicators the job is fulfilled
4. **Out-of-Scope Jobs** — jobs explicitly rejected or deferred
5. **Coverage Matrix** — table mapping each FR/NFR to the job(s) it supports

#### Example

```
/spec jtbd websocket  
/spec jtbd websocket --force  
/spec jtbd auth-v2 --agent architect
```

The `--force` flag overwrites an existing `JTBD.md`. The `--agent <name>` flag dispatches the analysis to a specific agent instead of the default explore agent.

---

## 15. Production feedback

`/spec feedback` appends advisory notes to `FEEDBACK.md`. These notes are surfaced during `/spec plan` regeneration — the agent is instructed to consider each note and add or adjust tasks to address identified issues.

### Workflow

1. Capture a production issue or insight:

```
/spec feedback websocket Users report intermittent disconnections when switching networks
```

2. During the next plan regeneration, the feedback is surfaced:

```
/spec plan websocket Use tokio-tungstenite with automatic reconnection and connection state track
```

The agent receives the `FEEDBACK.md` content and is instructed to address each note by adding or adjusting tasks.

### Configuration

The `sdd.feedback_loop` flag enables this feature:

```
{
  "sdd": {
    "feedback_loop": true
  }
}
```

---

## 16. End-to-end examples

### Example 1: Full SDD workflow for a new feature

# Step 1: Specify the feature

```
/spec specify websocket Add a real-time collaborative editing feature using WebSockets
```

# Step 2: Review the generated `SPEC.md`, resolve `[NEEDS CLARIFICATION]` markers

# (manually edit `specs/websocket/SPEC.md`)

# Step 3: Validate

```
/spec validate websocket
```

# Step 4: Generate the plan with technology context

```
/spec plan websocket Use tokio-tungstenite for WebSocket server, Redis for pub/sub
```

# Step 5: Generate task list and quickstart

```
/spec tasks websocket
```

# Step 6: Check coverage

```
/spec coverage websocket
```



```
# Step 7: Move through the lifecycle
/spec status websocket in_review
/spec status websocket approved

# Step 8: Preview the implementation plan
/spec impl websocket --dry-run

# Step 9: Implement
/spec impl websocket

# Step 10: Verify and archive
/spec status websocket implemented
/spec status websocket verified
/spec status websocket archived
```

### **Example 2: Quick-create for a small feature**

```
# Create everything in one step
/spec create csv-export Add CSV export for all list views

# Validate
/spec validate csv-export

# Implement
/spec impl csv-export
```

### **Example 3: Incremental feature addition**

```
# Add a new feature to an existing spec
/spec add websocket Add presence indicators showing who is currently viewing each document

# Validate the updated spec
/spec validate websocket

# Regenerate the plan from the updated SPEC.md
/spec update websocket

# Implement the new tasks
/spec impl websocket --task T-008
```

### **Example 4: Research-driven spec**

```
# First, run research on the topic
/research create oauth-study "OAuth2 PKCE flow for desktop applications"

# Then create a spec linked to the research
/spec specify auth-v2 Implement OAuth2 with PKCE --from-research oauth-study

# Validate, plan, and implement
/spec validate auth-v2
```

```
/spec plan auth-v2 Use oauth2 crate with PKCE, store tokens in SQLite
/spec impl auth-v2
```

#### **Example 5: JTBD analysis of an existing spec**

```
# Analyze the jobs a spec is hired to do
/spec jtbd websocket
```

```
# Review the generated JTBD.md
# (manually read specs/websocket/JTBD.md)
```

```
# Force regeneration after spec changes
/spec jtbd websocket --force
```

#### **Example 6: Production feedback loop**

```
# Capture a production issue
/spec feedback websocket Users report intermittent disconnections when switching networks
```

```
# Regenerate the plan with feedback surfaced
/spec plan websocket Use tokio-tungstenite with automatic reconnection
```

```
# The new plan will include tasks addressing the feedback note
/spec impl websocket
```

#### **Example 7: Filtering and searching specs**

```
# List all approved specs
/spec list --status approved
```

```
# Find specs about authentication
/spec search authentication
```

```
# List all specs including archived
/spec list --all
```

```
# Filter by ID prefix
/spec list --prefix auth
```

#### **Example 8: Task management**

```
# List all tasks in a spec
/spec task websocket
```

```
# Mark a task as in progress
/spec task websocket T-003 in_progress
```

```
# Mark a task as completed
/spec task websocket T-003 completed
```

```
# Mark a task as blocked
/spec task websocket T-005 blocked

# Implement a single task with its dependencies
/spec impl websocket --task T-005
```

---

## 17. Tips for good specs

- **Use specific feature descriptions.** “Add a real-time collaborative editing feature using WebSockets with OT conflict resolution” is better than “Add editing”.
  - **Use the SDD workflow for complex features.** Separate specification from planning so you can review and clarify requirements before committing to a technology approach.
  - **Resolve all clarification markers before planning.** Run `/spec validate` after editing `SPEC.md` to confirm no markers remain.
  - **Provide meaningful technology context.** `/spec plan` produces better plans when the tech context specifies crates, patterns, and architecture decisions.
  - **Link specs to research.** Use `--from-research` to ground specs in prior research, ensuring the specification reflects findings.
  - **Use `/spec add` for incremental features.** Don’t recreate a spec when adding a feature — use `/spec add` to extend it without disrupting existing requirements.
  - **Capture production feedback.** Use `/spec feedback` to record issues and insights that should inform future plan iterations.
  - **Run `/spec impl --dry-run` first.** Preview the execution order before committing to implementation.
  - **Check coverage before implementing.** `/spec coverage` reveals requirements without linked tasks.
  - **Use branch-per-spec for isolation.** Enable `sdd.branch_per_spec` to isolate spec work on a dedicated branch.
  - **Validate early and often.** Run `/spec validate` after any spec edit to catch issues before they propagate.
- 

## 18. LLM tool surface

Specs are also accessible to the agent as first-class LLM tools, so a coding agent can read and track specs without slash commands:

| Tool                          | Purpose                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------|
| <code>spec_read</code>        | Read a specification by ID (full <code>SPEC.md</code> content, requirements, tasks, status) |
| <code>spec_list</code>        | List specifications, optionally filtered by status                                          |
| <code>spec_search</code>      | Keyword search across spec titles and content                                               |
| <code>spec_task_update</code> | Update a task’s status (pending/in_progress/completed/blocked) within a spec                |

spec\_coverage

Generate a requirement-coverage report showing which requirements are linked to completed tasks

---

## 19. Troubleshooting

| Symptom                                    | Likely cause                                      | What to do                                                                  |
|--------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------|
| unknown subcommand                         | Typo or missing arguments                         | Run <code>/spec help</code> for the command reference                       |
| Spec not found                             | Invalid spec ID or spec doesn't exist             | Run <code>/spec list</code> to see available specs                          |
| invalid transition                         | Trying to skip a lifecycle stage                  | Follow the transition graph in section 6                                    |
| blocked by clarification markers           | Unresolved [NEEDS CLARIFICATION] markers          | Edit <code>SPEC.md</code> to resolve markers, then re-validate              |
| blocked by unchecked gates                 | Phase -1 gate checkboxes not checked              | Edit <code>PLAN.md</code> to check all gate checkboxes                      |
| no tasks found                             | <code>PLAN.md</code> missing or has no task table | Run <code>/spec plan</code> or <code>/spec create</code> to generate a plan |
| <code>/spec impl</code> produces no output | No pending tasks or all tasks completed           | Run <code>/spec impl --dry-run</code> to check task status                  |
| Branch not created                         | Not a git repo or branch already exists           | The spec is still created; check <code>git branch</code>                    |
| <code>data-model.md</code> not generated   | <code>sdd.data_model</code> not enabled           | Enable the flag in <code>ragent.json</code>                                 |
| <code>contracts/</code> not generated      | <code>sdd.contracts</code> not enabled            | Enable the flag in <code>ragent.json</code>                                 |
| JTBD overwrites existing file              | <code>--force</code> not specified                | Re-run with <code>--force</code> to overwrite                               |
| Feedback not surfaced in plan              | <code>sdd.feedback_loop</code> not enabled        | Enable the flag in <code>ragent.json</code>                                 |

---

*End of manual.*